

Surface Type

EPI Structure

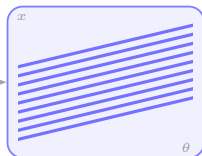
Cone Constraint

Depth Result

EPI Assumption Holds

Lambertian
(diffuse)

Uniform BRDF



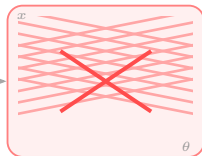
Linear structure

Slope = d (depth) $\frac{\partial I}{\partial \theta}$ PSNR: 32.4 dB SSIM: 0.93 **Accurate**

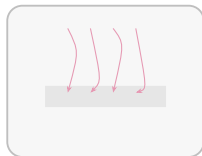
EPI Assumption Fails

Specular
(reflective)

Specular BRDF



X-shaped pattern

Dual slopes $\nabla_{\theta} I \neq$ PSNR: 22.1 dB SSIM: 0.78 $\Delta -10.3$ dBScattering
(translucent)

Subsurface scattering



Scattered pattern

Non-linear transport $\nabla_{\theta} I \neq$ PSNR: 21.8 dB SSIM: 0.76 $\Delta -10.6$ dB

Key insight: Lambertian assumption $\frac{\partial I}{\partial \theta} = d \cdot \frac{\partial I}{\partial x}$ is violated by non-Lambertian BRDF, causing ≥ 10 dB PSNR drop.